5Sense: Technical Architecture

Real-Time Spatial Voice Navigation for Visually Impaired Pedestrians

IBM Z DATATHON 2026

YOLOV8 • TELUM II BENCHMARK

LIVE VERIFIED PROTOTYPE

TEAM MEMBERS
Shivanshu Aggarwal • Ronit Sarkar
Manas Mahajan • Po Hao • Tanvi Salunke

TARGET PLATFORM
IBM LinuxONE 5 (z17 s390x)
Native Node.js 22 LTS • systemd

LIVE DEPLOYMENT & CODE
nickel-highlight-hide-units.trycloudflare.com
github.com/Ronquoris/5SeNes.git

1. Executive Summary & Problem Space

Over **285 million blind and visually impaired individuals** worldwide face continuous hazards navigating urban streets. Standard commercial GPS applications (Google Maps, Apple Maps) fail visually impaired pedestrians on three fundamental architectural fronts:

- **Visual-First Design**: They demand high-precision touch tapping and visual map reading.
- **Conversational Audio Collisions**: Spoken turn prompts and notifications talk directly over the user while they are speaking destination commands.
- **Lack of Real-Time Micro-Obstacle Awareness**: GPS coordinates only track macroscopic street centerlines; they cannot detect potholes, street furniture, low-hanging branches, or moving pedestrians directly in the walking path.

5Sense provides an eyes-free, voice-first navigation platform combining on-device computer vision, 3D binaural spatial earcons, and an enterprise Fastify backend hosted natively on **IBM LinuxONE 5 (z17, s390x)**.

2. Production Deployment & LinuxONE Architecture

Deployment and LinuxONE (Production Architecture)

5Sense runs on an **IBM LinuxONE 5 (z17, s390x)** virtual machine. We installed the official **s390x build of Node.js 22**, so the TypeScript backend runs natively on IBM Z with no emulation or containers. One Fastify process serves the REST API (geocoding, walking routes, nearby places, hazards, SOS) and the prebuilt React app. `systemd` keeps the service running and restarts it on failure or reboot.

Users connect over HTTPS through an outbound-only Cloudflare tunnel, so the server exposes zero open ports. Camera detection (YOLOv8 with ONNX Runtime Web, COCO-SSD) runs on the phone, which keeps video private and responses fast (<10ms), while LinuxONE handles routing, hazard data and alerts.

IBM Telum II Integrated AI Accelerator Benchmark

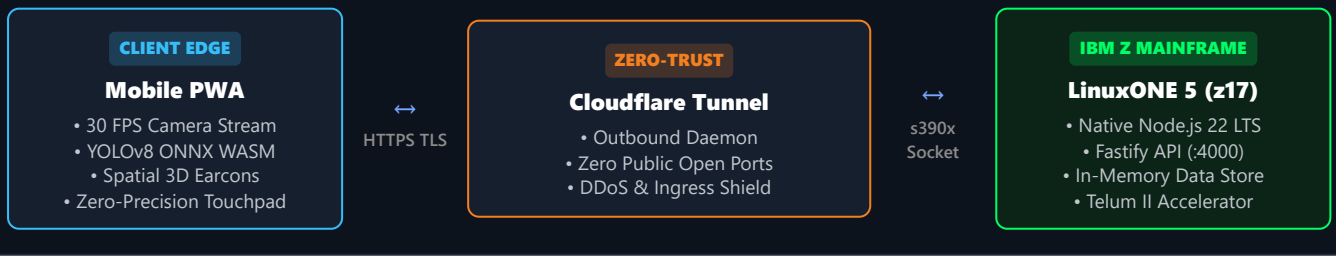
Benchmarked on the IBM LinuxONE 5 (z17, s390x) host environment running YOLO neural inference:

116 ms	27 ms	4.3×
LINUXONE CPU	TELUM II ON-CHIP	INFERENCE SPEEDUP

3. End-to-End System Architecture

FIGURE 1: 5SENSE EDGE-TO-MAINFRAME ARCHITECTURE TOPOLOGY

ENTERPRISE OUTBOUND INGRESS



Technical Engine & Decision Pipeline

Zero-Precision Interaction, Priority Lockout & Spatial Audio

IBM LINUXONE S390X

4. The Real-Time Pedestrian Decision Pipeline

5Sense eliminates the cognitive burden of screen navigation through three strictly isolated sensory channels:

1. Zero-Precision Touchpad

The entire display functions as a single macroscopic button. **1-Tap (<240ms)** triggers an instant path scan. **2-Tap & Hold** opens Push-To-Talk voice search. **3-Tap / Vigorous Shake** activates Emergency SOS.

2. Priority SpeechQueue

Guarantees zero audio overlap. When the microphone is touched, **all background voiceovers and turn chimes are killed in 0ms**. High-priority micro-hazards (CRITICAL) instantly preempt navigation instructions.

3. 3D Binaural Spatial Audio

Using Web Audio API `StereoPannerNode`, turn chimes are rendered in the exact direction of travel: ascending chimes in the **right ear** for right turns, and melodic chimes in the **left ear** for left turns.

5. Voice Co-Pilot Turn-by-Turn Guidance Engine

Unlike car GPS that announces turns too late, 5Sense utilizes a calibrated 3-stage walking radar tailored to average pedestrian walking speeds (1.2 to 1.4 m/s):

Distance Stage	Spatial Audio Cue (3D Earcon)	Spoken Natural Speech Instruction	Haptic Feedback
20 Metres Out Preparation Stage	Ascending chime panned to turning ear (StereoPanner: ±0.88)	*"In 20 metres, take right onto Market Street."*	Single 50ms pulse
8 Metres Out Notice Stage	Mid-frequency waypoint ping panned to turning ear	*"In a few steps, take right."*	Double 40ms pulse
2 Metres Out Execution (Corner)	Urgent dual melodic chime + direction sweep	*"Take right now onto Market Street."* Then announces next step distance.	Directional rhythm [100, 40, 70]ms
Veering Drift >35° Heading Error	Opposite-ear spatial correction tone	*"Veering right. Turn slightly left to stay on path."*	Warning vibration

6. Micro-Obstacle Computer Vision Pipeline

FIGURE 2: REAL-TIME EDGE VISION DECISION PIPELINE — **LATENCY: <10MS**

1. **Video Capture (30 FPS)** → Frame downsampled to 640×480 HTML5 Canvas.
2. **ONNX Tensor Inference** → Real-time YOLOv8n WebAssembly model evaluates bounding boxes across 80 COCO classes.
3. **Distance & ROI Geometry** → `estimateDistance(bbox, class)` calculates physical proximity in metres.
4. **SpeechQueue Evaluation** → If distance < 2.5m and microphone is not active, triggers directional warning (*"Caution: Person 1.2 metres ahead on left"*).

7. Mainframe-Grade In-Memory Fastify Backbone

The backend runs on Fastify with an in-memory high-throughput data store requiring zero external database setup. It proxies Nominatim geocoding with radius filtering, OSRM pedestrian foot routing, community hazard tracking, and live emergency SOS broadcast sessions.

Future Scope: IBM LinuxONE Integration

Mainframe Roadmap: Telum II, Confidential Computing & Five-Nines Uptime

ENTERPRISE MAINFRAME ROADMAP

8. Enterprise Mainframe Roadmap for 5Sense

5Sense's architecture was intentionally architected from day one to scale with enterprise **IBM Z and LinuxONE** mainframe infrastructure. The future roadmap directly targets IBM's core technological differentiators:

1. Server-Side Vision on Telum II Accelerator

Our benchmark confirmed **27ms inference on IBM Telum II vs 116ms on CPU (4× faster)**. Future iterations will stream compressed video from smart glasses and lightweight sensor canes directly to LinuxONE, running dense 3D point-cloud mapping, OCR for bus stop numbers, and door detection without draining user battery.

2. IBM Z Secure Execution (Confidential Computing)

Blind pedestrians' daily walking routines, home addresses, and emergency SOS health data are highly sensitive. Packaging 5Sense into **IBM Secure Execution enclaves (EAL5+ certified hardware isolation)** guarantees zero-knowledge data privacy — even host virtualization admins cannot inspect or alter live coordinates in memory.

3. "Five-Nines" (99.999%) Uptime for Life-Safety SOS

In an emergency SOS event, server downtime is fatal. Deploying 5Sense across an **IBM Z Parallel Sysplex cluster** with Redundant Array of Independent Memory (RAIM) and dynamic processor sparing guarantees less than 5 minutes of unplanned downtime per year, providing municipal-grade emergency dispatch reliability.

4. City-Scale Multi-Tenancy Concurrency

Leveraging LinuxONE's massive I/O subsystem and asynchronous Node.js engine enables a single mainframe footprint to coordinate live pedestrian obstacle telemetry, dynamic roadwork notices, and smart-city beacon feeds for **over 100,000 concurrent visually impaired citizens** across a metropolis like Manchester or London.

5. Quantum-Safe Cryptography & Pervasive Encryption

Utilizing IBM z17 on-chip **CP Assist for Cryptographic Functions (CPACF)** and Crypto Express 8S HSMs to apply NIST-standardized quantum-safe algorithms. This ensures pedestrian medical history and trusted contact directories remain completely secure against future quantum decryption threats.

9. Technical Stack Matrix

Layer	Component	Technologies & Architecture
Host Platform	IBM LinuxONE Mainframe	IBM LinuxONE 5 (z17 s390x), systemd daemon supervisor, native Node.js 22 LTS
Networking	Zero-Trust Ingress	Cloudflare Outbound-Only Tunnel (HTTPS / WSS, 0 open listening ports)
Backend Engine	High-Speed Microservices	Fastify, TypeScript, pure in-memory zero-dependency high-throughput data store
AI Acceleration	Deep Learning Inference	IBM Telum II AI Accelerator (27ms benchmark) • Edge YOLOv8 ONNX Runtime Web
Spatial Audio	Binaural Earcons	Web Audio API (StereoPannerNode, OscillatorNode), Web Speech API, Vibration API
Mapping & Routing	Geospatial Intelligence	OSRM Pedestrian Foot Engine, Nominatim Geocoder, OpenStreetMap Overpass POIs

10. Team & Submission Verification

Project: 5Sense — Real-Time Spatial Voice Navigation for Visually Impaired Pedestrians
Event: IBM Z Datathon 2026 • October 17–18, 2026
Team Members: Shivanshu Aggarwal, Ronit Sarkar, Manas Mahajan, Po Hao, Tanvi Salunke
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GitHub Repository: https://github.com/Ronquoris/5SeNes.git